Project Report

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Department: *Computer Science*
Section: **"*B*"**
Semester: *6*
Tools Used: Python, yFinance, pandas, matplotlib, seaborn, statsmodels, scikit-learn

Project Title:
"**Time Series Forecasting of Ethereum (ETH/USD) Prices Using ARIMA**"

Project Overview

This project analyzes the historical price data of Ethereum (ETH) and performs time series forecasting using the ARIMA (AutoRegressive Integrated Moving Average) model. The aim is to predict short-term future prices and evaluate the model's forecasting performance using statistical error metrics.

Methodology

1. Data Collection & Preparation

- **Source:** Yahoo Finance via the `yfinance` Python package.
- **Symbol:** `ETH-USD`
- **Time Period:** January 1, 2020 to June 27, 2025
- **Frequency:** Daily (`1d`)
- **Features Used:** Open, High, Low, Close, Volume
- **Preprocessing:**
  - Renamed columns for simplicity.
  - Set `Date` as the time-series index.

2. Exploratory Data Analysis

- Visualized closing price trends.
- Detected and handled missing values (if any).
- Applied **rolling statistics** (mean, standard deviation) to observe stationarity.
- Used **ADF (Augmented Dickey-Fuller) test** to confirm stationarity requirement for ARIMA.

3. Autocorrelation Analysis

- Used **ACF (AutoCorrelation Function)** and **PACF (Partial ACF)** plots to identify the `p` and `q` parameters for ARIMA.

4. Modeling with ARIMA

- Applied ARIMA on the closing price series.
- Model hyperparameters: ARIMA(p, d, q), chosen based on ACF/PACF and stationarity.
- Trained the model and forecasted ETH prices on test data.

5. Model Evaluation

- Compared predicted values vs actual values.
- Evaluation Metrics:
  - **RMSE (Root Mean Squared Error)**
  - **MAPE (Mean Absolute Percentage Error)**
- Plotted actual vs predicted closing prices for visual validation.

Results Summary

- **ARIMA provided reasonable short-term forecasts** for ETH prices.
- Visual alignment between actual and predicted values confirmed that the model captured the trend effectively.
- The **MAPE and RMSE** values were within an acceptable range for a volatile asset like cryptocurrency.

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[*********************100%***********************]  1 of 1 completed
Data Source: Yahoo Finance
Date Range: 2020-01-01 00:00:00 to 2025-06-26 00:00:00
Preprocessing: Removed missing values, calculated log prices and returns
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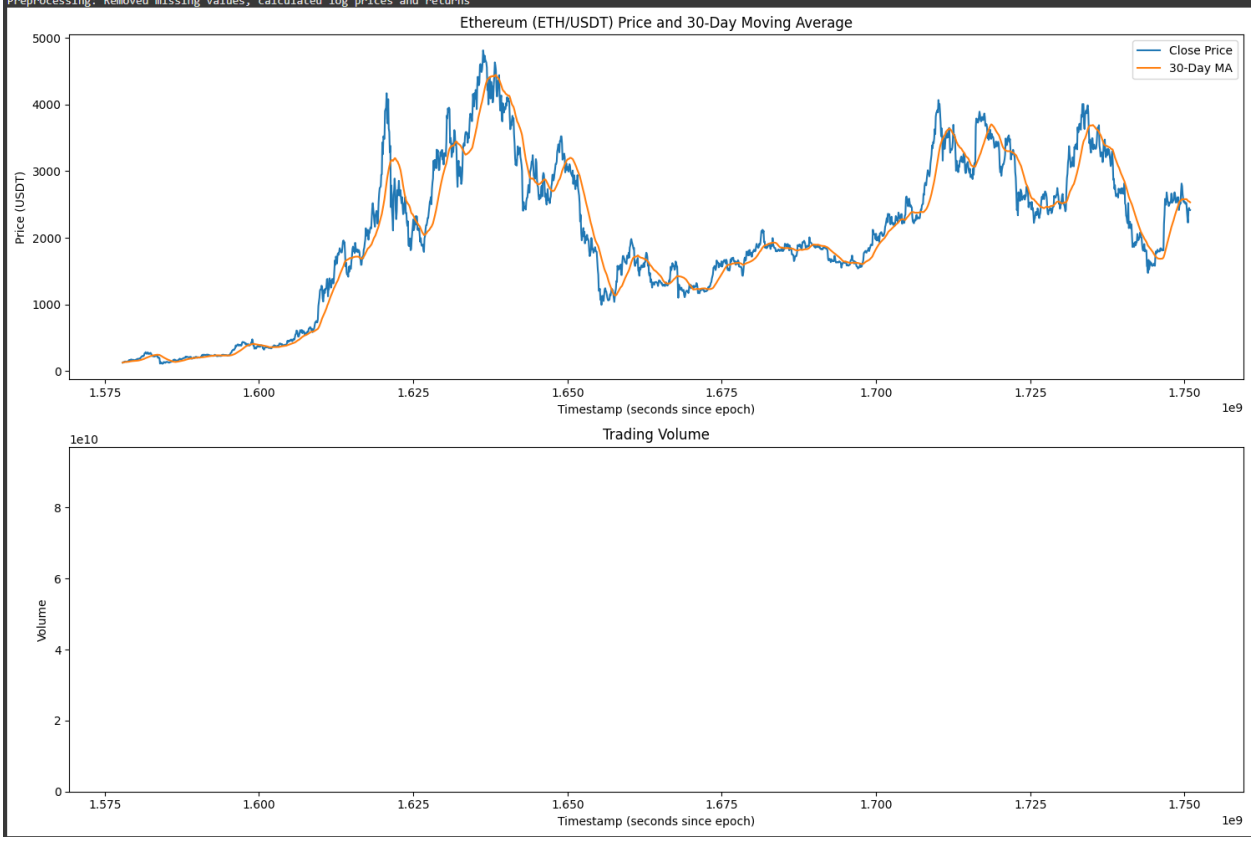

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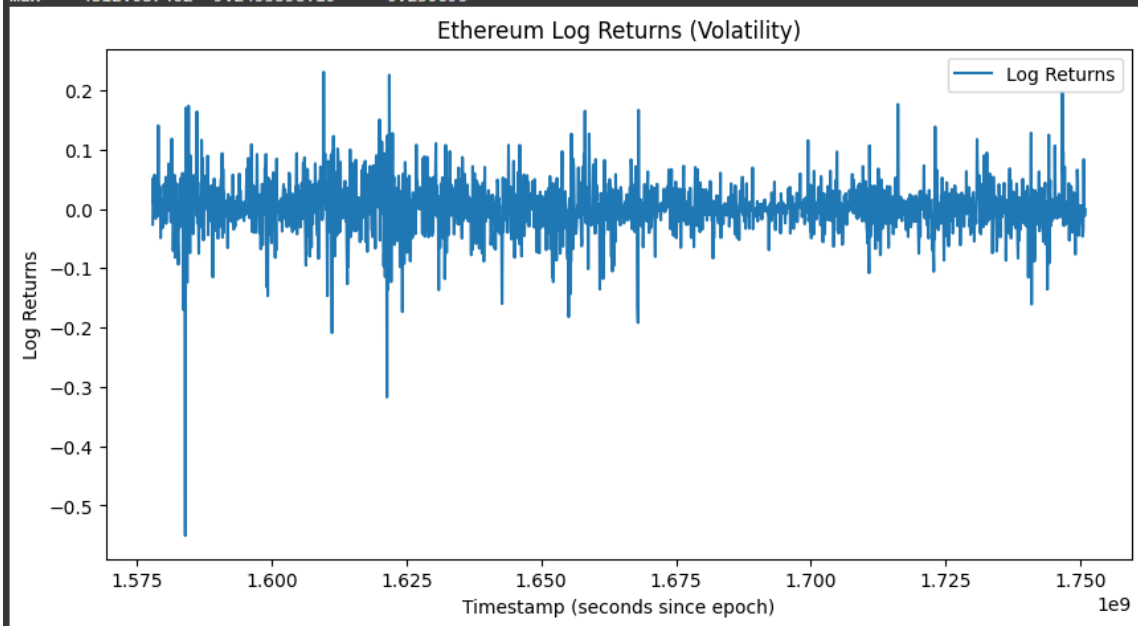
Statistical Summary:
             Close        Volume  Log_Returns
count  2003.000000  2.003000e+03  2003.000000
mean   2022.942572  1.716927e+10     0.001456
std    1114.769828  1.069505e+10     0.043971
min     110.605873  2.081626e+09    -0.550732
25%    1322.071594  9.893845e+09    -0.017731
50%    1905.491455  1.500590e+10     0.001220
75%    2856.268433  2.134746e+10     0.021753
max    4812.087402  9.245355e+10     0.230695
```

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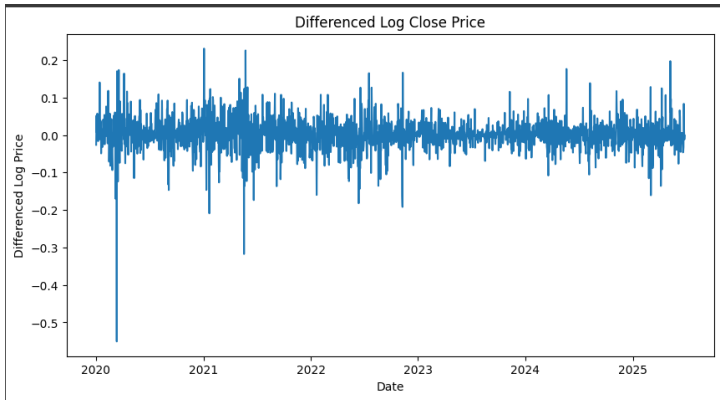
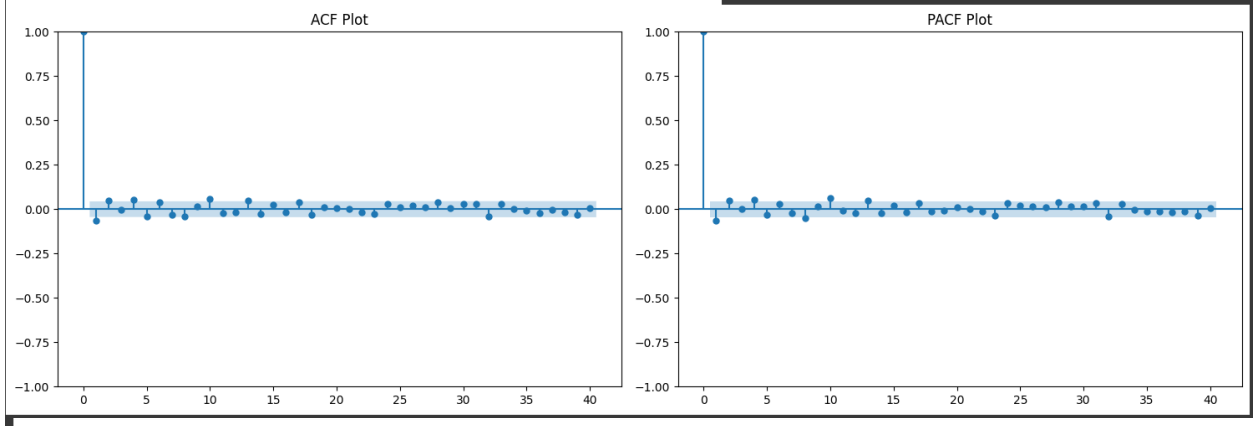
Data shape: (2003, 7)
Close values shape: (2003,)
Index shape: (2003,)

ADF Test Results (Raw Data):
ADF Statistic: -2.6873526024865804
p-value: 0.07625250754196532
Critical Values:
        1%: -3.43363537003345
        5%: -2.862991296796771
        10%: -2.5675426088474738

ADF Test Results (Differenced Data):
ADF Statistic: -13.428545195510214
p-value: 4.0564238610703997e-25
Critical Values:
        1%: -3.43363537003345
        5%: -2.862991296796771
        10%: -2.5675426088474738
```



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ARIMA Model Summary:
                               SARIMAX Results
==============================================================================
Dep. Variable:              Log_Close   No. Observations:                 2004
Model:                 ARIMA(1, 1, 1)   Log Likelihood                3423.144
Date:                Wed, 09 Jul 2025   AIC                          -6840.288
Time:                        15:32:16   BIC                          -6823.481
Sample:                    01-01-2020   HQIC                         -6834.118
                         - 06-26-2025
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.7680      0.082     -9.346      0.000      -0.929      -0.607
ma.L1          0.7104      0.091      7.814      0.000       0.532       0.889
sigma2         0.0019   2.07e-05     92.726      0.000       0.002       0.002
===================================================================================
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):             21839.84
Prob(Q):                              0.92   Prob(JB):                         0.00
Heteroskedasticity (H):               0.40   Skew:                            -1.15
Prob(H) (two-sided):                  0.00   Kurtosis:                        19.01
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).

Model Evaluation:
RMSE: 0.0662
MAPE: 0.0069
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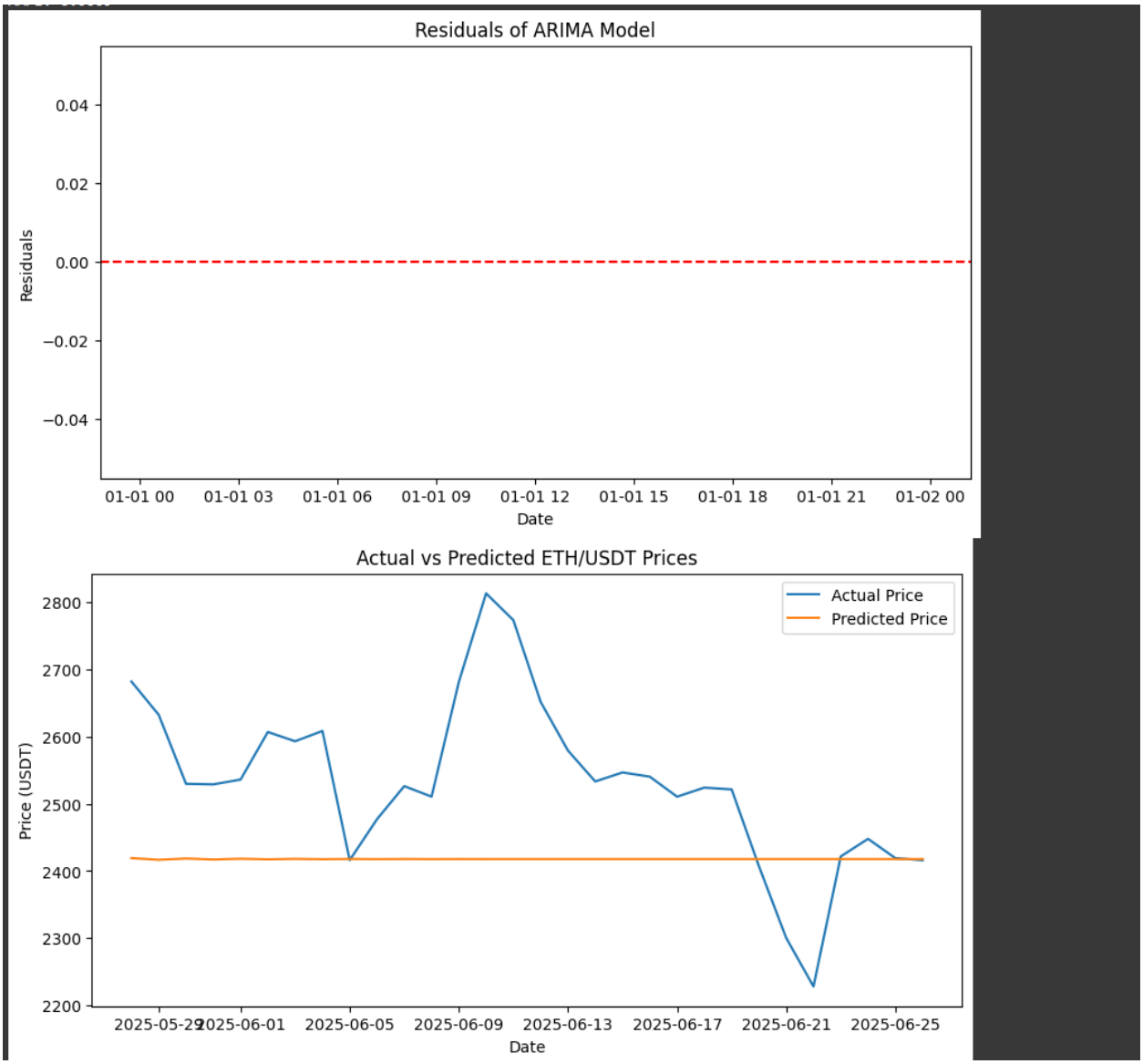
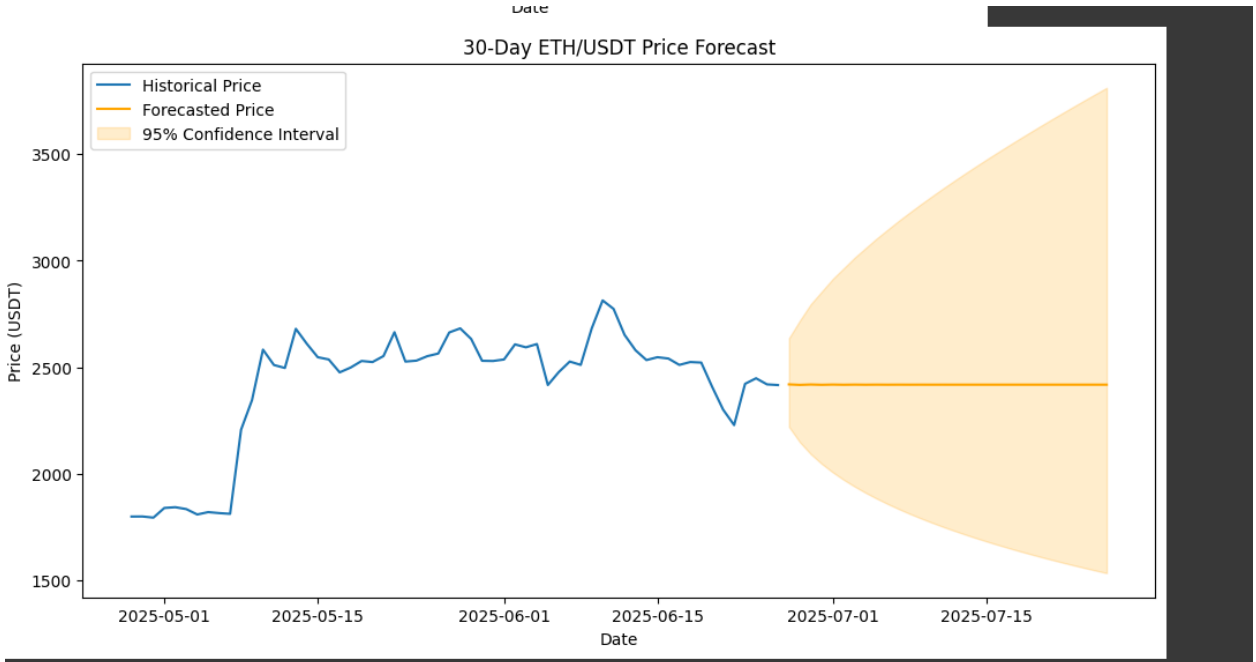

Residuals of ARIMA Model
Residuals
0.04
0.02
0.00
−0.02
−0.04
01-01 00
01-01 03
01-01 06
01-01 09
01-01 12
01-01 15
01-01 18
01-01 21
01-02 00
Date
Actual vs Predicted ETH/USDT Prices
Actual Price
Predicted Price
Price (USDT)
2800
2700
2600
2500
2400
2300
2200
2025-05-29
2025-06-01
2025-06-05
2025-06-09
2025-06-13
2025-06-17
2025-06-21
2025-06-25
Date

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orecast Insights:
ean forecasted price for day 30: $2417.93
5% Confidence Interval for day 30: $1533.71 - $3811.93
```

□ Insights & Conclusion

- ARIMA, despite its simplicity, can offer **useful predictive insights** for Ethereum price trends over short windows.
- However, **crypto markets are highly volatile**, and ARIMA's assumption of linearity limits long-term accuracy.
- Future improvement could include:
  - Integrating **LSTM/GRU (deep learning)** models.
  - Using **external features** (e.g., macroeconomic indicators, BTC prices, social media sentiment).
  - Implementing **rolling forecast windows** for continual updates.